

Tactile Models for the Visualization, Conceptualization, and Review of Intermolecular Forces in the College Chemistry Classroom

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ABSTRACT: The authors describe the construction and use of tactile models for demonstrating intermolecular forces. These models are composed of inexpensive materials and can be used in college chemistry classrooms of varying levels including general and organic chemistry. The models were designed to be paired with an accompanying activity to encourage students to compare and contrast the three types of intermolecular forces (hydrogen bonding, dipole–dipole forces, and London dispersion forces) as well as determine the relationship between intermolecular forces and the physical properties of specific molecules. More importantly, the handheld models allow for visualization of molecular interactions to promote multisensory learning. In this work, the authors discuss the development, implementation, and survey results of a modeling activity for conceptualization and review of intermolecular forces in introductory, general, and organic chemistry classrooms.

KEYWORDS: *First-Year Undergraduate/General, Second-Year Undergraduate, General Public, Hydrogen Bonding, Organic Chemistry, Demonstrations, Hands-On Learning/Manipulatives, Molecular Properties/Structure*

INTRODUCTION

Intermolecular forces (IMFs) are a foundational concept for understanding phases, reactivity of functional groups, and even some biological functions taught in general and organic chemistry courses.^{1–3} Although a variety of teaching techniques and laboratory experiments have been developed to demonstrate the relationship between physical properties of compounds, such as boiling point and their intermolecular forces,^{4–7} students often fail to connect these concepts to what is occurring on the molecular level in terms of the consequential three-dimensional arrangements of the interacting molecules.⁸ Indeed, work by Nicoll suggests that students often fail to develop mental models of interacting molecules.⁹ Students may also misconceive the factors that affect the strength of IMFs as well as confuse *intermolecular* forces with *intramolecular* forces.^{8,10,11} Cooper et al. have demonstrated that students often can provide textbook level definitions but leave general chemistry and enter into organic chemistry without a thorough conceptual understanding of IMFs.^{8,12,13} In particular, students struggling with a conceptual understanding of the differences between covalent and noncovalent bonds are presented with various drawing interventions and assessments such as those used in the Chemistry, Life, The Universe and Everything curriculum (CLUE). These drawings reveal students' ability to provide definitions, but this does not necessarily translate to a 2D or 3D understanding. This can persist through to subsequent courses. Henderleiter et al. have shown that organic chemistry students still struggle with understanding IMFs and that when students solely memorize the terms, they have an incomplete or incorrect understanding of hydrogen bonding.¹⁴

Using Modeling to Teach Intermolecular Forces

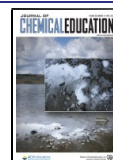
When first introduced to intermolecular forces (IMFs) either in lecture or in an activity, a student may form misconceptions based on prior knowledge or their perceptions of the macroscopic world. To connect macroscopic perceptions to a microscopic understanding of IMFs, we have developed handheld models to foster 3D visualization of intermolecular forces and to promote encoding of new concepts into memory via tactile learning.^{15,16} These models attempt to address the differentiation of covalent bonds vs IMFs, strength of the IMF attractions, the significance of the shape of the molecule as it relates to its impact on the ability to form attractions, and the collective strength of these attractions in a molecule regardless of the relative strengths of the different types of IMFs being modeled and compared (e.g., London dispersion versus hydrogen bonding). The models are intended not to be a replacement for more in-depth discussion of the concepts but to provide a 3D representation to aid students' visualization of the interactions of the molecules.

Modeling and models are a core foundation of understanding science, and the ability to translate between 2D and 3D is vital in science.¹⁷ Physical models allow students a tangible way to view and manipulate structures three dimensionally. Studies have shown students prefer handheld or tangible models when learning concepts.^{18,19} It is known that students as novices have difficulty transitioning between

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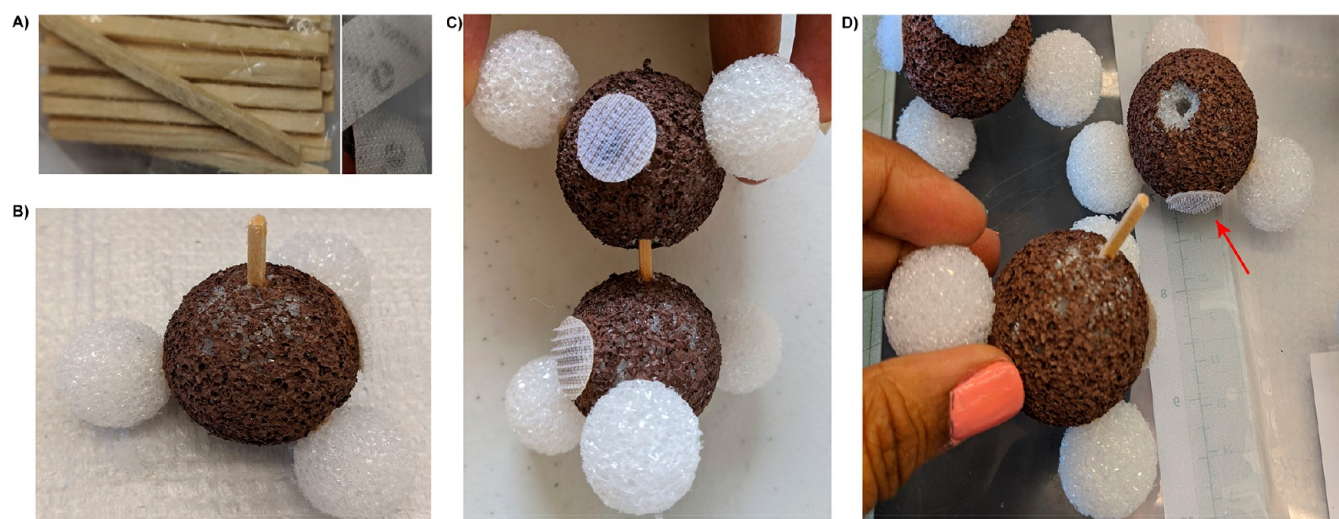


Figure 1. Model construction of ethane. (A) Wooden dowels and Velcro for model construction. (B) Placement of wooden dowel in Styrofoam ball representative of a carbon atom for assembly to another Styrofoam ball; super glue is applied to the surface of the Styrofoam ball surrounding the wooden dowel. (C) Assembly of Styrofoam balls to represent the ethane molecule. (D) Assembly components of the ethane molecule. The red arrow denotes Velcro attachment necessary for demonstrating London dispersion forces.

the microscopic, macroscopic, and symbolic nature of chemistry. As experts, chemists easily slip between the points of Johnstone's triangle of the three basic components of chemistry (macro, micro, and representative nature).²⁰ Evaluation of students' discussions in small groups by Becker et al. revealed that students focused heavily on the symbolic nature, even though they had the opportunity to relate the three levels.²¹ One way to potentially bridge the gap for students is through the use of tangible models. Molecular models can provide a method for offloading cognition onto an object as well as support how new concepts are encoded into memory through physical manipulation of the model.²¹ Padalkar and Hegarty have shown that an experimental group of students using organic ball-and-stick models performed better than a control group when asked to use models to check their answers in a pretest on stereochemistry concepts.¹⁵ Anastopoulou et al. suggest that multimodal interactions can allow transition between concrete and abstract concepts.²² They suggest that this physical manipulation can lead to better understanding of science concepts.

The models developed in this work utilized Styrofoam balls and Velcro to simulate IMFs. Although similar models to demonstrate hydrogen-bonding have been reported previously,²³ handheld models demonstrating the different types of intermolecular forces are presented here. Handheld models have been shown to improve student learning of chemical concepts such as stereochemistry,¹⁵ macromolecular structures,^{18,24} and IR spectroscopy.²⁵ In addition, it has been suggested that physical manipulation can support understanding of abstract concepts among students.²² Thus, we designed tangible models and implemented a modeling activity for the chemistry classroom for the purposes of teaching and reviewing intermolecular forces. Though computer-generated models can add to learning concepts as seen in Harris et al.,¹⁸ as mentioned by Roberts et al., simulations of 2D models require students to make inferences on the 3D nature of molecules beyond what can be demonstrated in 2D simulations. Their work suggests that although computer-generated models are useful, it appears that reducing the cognitive load associated with making

correlations between 2D images and 3D molecules may be alleviated with tangible models as the initial exposure. Our models provide one less barrier to students in understanding how molecules interact, through a more tangible as well as visible approach. This may allow students to more easily form mental models, as this was an issue noted by Nicoll,⁹ and potentially allow students to have deeper conceptually driven discussions on IMFs with their instructors, rather than simply to memorize the definitions as found by Henderleiter.¹⁴

■ MODEL CONSTRUCTION

The IMF models were constructed from materials that an instructor can easily access. Though 3D printed models may also be made and manipulated, this format gives an example of easily accessible craft-store materials. The models are made of polystyrene foam balls, such as Styrofoam balls, of varying sizes and hook and loop fasteners, such as Velcro,²⁶ of three different strengths. The Styrofoam balls, wood dowels, metal wiring, and Velcro were purchased from a local craft store (Figure 1A–C). The Styrofoam balls were painted simply for differentiation purposes. Velcro was added to provide the simulation of strengths (Figure 1D). The strongest Velcro was used for the molecules that participated in hydrogen bonding, the medium strength Velcro for the molecules that participate in dipole–dipole forces, and the weakest Velcro for the molecules that only participate in London dispersion forces. The models were assembled with super glue, wooden dowels, and wire to reinforce the models to represent the covalent bonding (Figure 1). The Velcro attraction is used to represent the IMFs (weak attraction), as compared to the glued Styrofoam atoms that are used to represent covalent bonding.

Figures 2–4 show the design, interactions, and arrangement of the models. Figure 3 shows an example of the interactions between models. The models were given a reasonable simulation of the actual shape of the molecules. Due to the limitation of sizes of Styrofoam balls, the authors attempted to vary the sizes to show variability in the “atoms” as best as possible (Figure 4). The “atoms” were painted to allow students to be able to identify the atoms in the model. The different molecules are represented and grouped in three

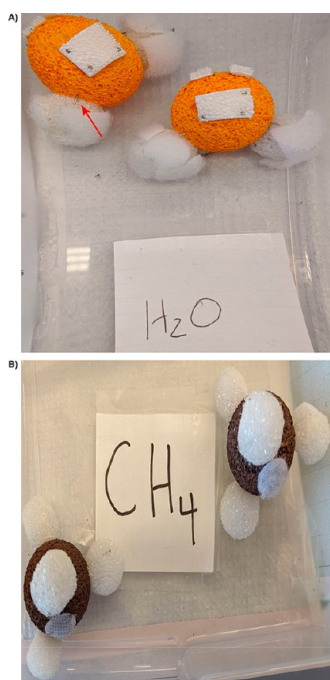


Figure 2. Close-up of a few representative models. (A) Water molecule models; the red arrow denotes wire necessary to reinforce models with high-strength Velcro to represent H-bonding. (B) Methane molecule models with single low-strength Velcro sticker to represent London dispersion forces.

different boxes according to the type of intermolecular force. Table 1 shows the identity of the representative molecules in each box; there were two models of each molecule in each box (i.e., in Box 1, there were two NH_3 models, two H_2O models, etc.). Figure 4 is an image of how molecules were placed in the box. However, each box may contain one molecule that “does not belong”, allowing students to identify the intermolecular force corresponding to this molecule.

MODELING ACTIVITY AND IMPLEMENTATION

An activity was designed to pair with the models to encourage students to interact with these models and articulate the IMFs that the models represent. There are three parts to the modeling exercise: (1) preactivity information and assessment, (2) modeling with activity instructions and questions, and (3) application questions and information. All student handouts and instructor notes are in the [Supporting Information](#).

The preactivity consisted of questions asking students to define IMFs, identify factors that affect IMFs, and describe what the students know about hydrogen bonding (H-bonding), dipole–dipole forces, and London dispersion forces. Students were just graded on submission to encourage them to write on what they remembered or read. Students’ responses to the questions were evaluated as advanced, intermediate, basic, or inadequate. Advanced statements were those that incorporated most of the completely correct explanation of the terms with some explanation of their meanings. Their answers seemed mostly adequate as compared to the other students. The intermediate answers were those that provided mostly sufficient information or a more detailed definition but may contain an error in the explanation or definition. Basic answers were those that gave mostly correct definitions (textbook type or their own) but with minimal information. Inadequate responses were those that ranged from major errors in explanations or insufficient definitions to not answering the question asked or saying they did not remember. Samples of each type of preactivity answer are presented below:

- *Advanced* response to “Define IMFs”: “They are forces, often referred to as ‘bonds’, between (not within) molecules. As the prefix -inter suggests (like interstate), these forces act across the distances between molecules rather than their individual atoms, and they are almost always weaker than intramolecular bonds such as ionic and covalent. These forces are the result of differences in partial charges of atoms in molecules resulting from the relative concentration of electrons, referred to

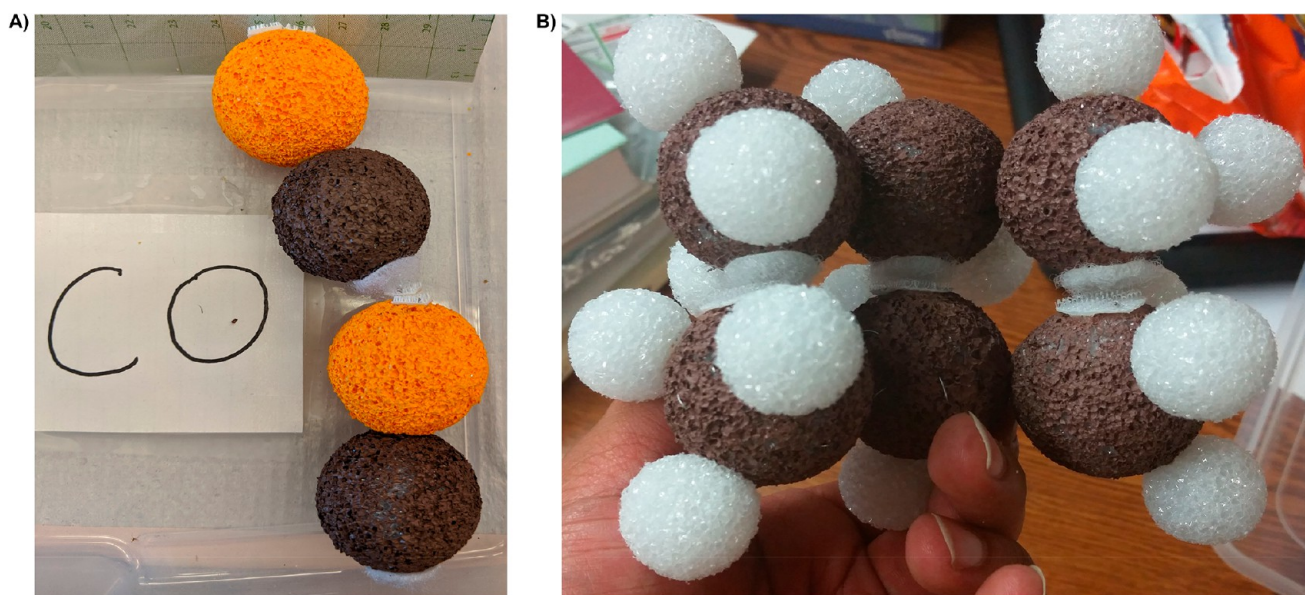


Figure 3. Examples of modeling intermolecular forces (IMFs) using the components of the Velcro’s hooks and loops. (A) Attachment of two CO models to demonstrate dipole–dipole forces. (B) Attachment of two propane (C_3H_8) models to demonstrate London dispersion forces.



Figure 4. Model layout showing individual boxes with both the cover on and the cover off. (A) Layout of the models in Box 1. (B) Layout of models in Box 2. (C) Layout of models in Box 3. Hydroxyacetone was initially used as a misfit molecule but was removed in later implementations.

primarily as dipoles, or in more extreme cases as polarity.”

- *Intermediate* response to “What do you know about H-Bond”: “A hydrogen bond is a rather strong bond and

can only occur between hydrogen and nitrogen, oxygen, fluorine, or carbon.”

- *Basic* response to “Define IMFs”: “Interactions between molecules.”

Table 1. Represented Molecules in Each Box

Box 1	Box 2	Box 3
H ₂ O	CH ₃ Cl	C ₂ H ₆
NH ₃	SO ₂	C ₃ H ₈
CH ₃ OH	CO	C ₅ H ₁₂
H ₂ S	CH ₄	C ₃ H ₆ O ₂ ^a

^aMolecule may be removed from the box and is shown removed in Figure 4C.

- *Inadequate* response to “What do you know about H-Bond”: “Weak bond between hydrogen atoms.”

Students handled the models and completed the corresponding activity in class in groups. The format of the activity was designed to facilitate instructors in evaluating how the models impacted the application questions as well as to minimize the time students needed the model kits. It also allowed the instructors to be able to circulate better and troubleshoot any problems students had with the boxes. For larger classes, half of the student groups started the assignment using the models (Activity A), and another related assignment was developed for the other half of the groups until a box became available (Activity B); one group of students would handle the models while the other group of students would work on the application questions, and then, the focus of the two groups (modeling versus application questions) would switch.

The three boxes were rotated throughout these student groups to allow each student the opportunity to work with the models from each box. While handling the models, students were asked to identify the molecule with the weakest IMF and the strongest IMF in each box. In addition, students were asked to identify the types of IMFs represented by the molecules in each box and the molecule that “did not belong”. The molecules that “did not belong” were placed as a way for instructors to observe students’ ability to differentiate the IMF strength. The application questions asked the students to evaluate the boiling point data for all the molecules in the activity and correlate these to the intermolecular forces. Students in both groups completed some of these questions in-class and finished the application questions at home after completing the in-class modeling activity.

When evaluating students’ descriptions of the boiling point trends and relationships, student responses were characterized as advanced, intermediate, basic, or inadequate just as before in the preactivity; however, instructors communicated to students that they anticipated more complete and accurate responses because students had received more information in-class throughout the activity. Thus, the student responses were also categorized with these higher expectations in mind. Advanced students included an evaluation of the shape of the molecule, number and accessibility of lone pair electrons, and discussed unexpected trends. A basic response may have been correct but was more like a textbook statement such as “H-bonding causes higher boiling points due to the higher strength of the force, making it less willing to break when heated.” A summary of students’ responses are in the [Supporting Information](#).

Student Participants

The activity was implemented in several college chemistry classrooms including classes in Introductory Chemistry (includes topics in general, organic, and biochemistry),

Organic Chemistry I, Organic Chemistry II, and General Chemistry II. The activity was used to strengthen student understanding of IMFs at a microscopic level as a new concept (in Introductory Chemistry and General Chemistry II) or as a review tool for advanced students (Organic Chemistry I and II). The number of students that participated in this activity is shown in Table 2. The Organic Chemistry II

Table 2. Number of Student Participants from Each Course Level

Class	N
Introductory Chemistry (IC)	11
General Chemistry II (GCII)	35
Organic I (OCI)	46
Organic II (OCII)	33

and Introductory Chemistry classes were our pilot group, and the General Chemistry II and Organic Chemistry I classes were given the activity the following semester.

Model and Survey Results

With the models, students were able to examine the difference in strengths of the IMFs and visually observe how the molecules could be interacting. About 3/4 of students classified the strengths in each box correctly, with box 3 almost 90% correct. Students appeared to begin applying the concepts, but instructors are encouraged to follow up with additional exercises or homework to extend students’ understanding of IMFs beyond the visual aspects of the models. It may be helpful to have questions in which students redefine the IMFs to observe changes in their definitions compared to the preactivity. More details on our students’ responses are in the [Supporting Information](#).

To assess student perceptions of the modeling activity, we distributed a brief survey to the students after the postactivity. Survey responses are summarized in Table 3. This table shows

Table 3. Survey Responses Represented as Percentages of Student Responses under Each Category

Prompts for Student Response	Students Selecting Response Categories, % (N = 125)		
	Positive	Neutral	Negative
How easy are instructions to understand?	61.5	28.2	10.3
Enjoy activity?	51.4	32.4	16.2
Activity helpful?	76.3	0.0	23.7
Visualization of interactions?	72.0	19.0	9.0

that we received mostly positive feedback from the students. Most of the students found the activity enjoyable and helpful. This positive feedback shows that most students were willing to do this activity in a classroom setting. Complaints are centered on the limited number of boxes or time with the boxes. Additionally, Organic Chemistry I students more commonly had a neutral response to the activity; student explanations for this neutral response included requests for more problems or statements that they had already learned IMFs and did not require review of the concept.

Limitations and Suggestions for the Activity

The intention of our models was to reduce the barrier associated with visualization of the molecules interacting. However, there are a few limitations to using these and any

model. One limitation associated with this activity is the durability of the models. Students should be instructed at the beginning of the activity to be gentle with the models; vigorously and forcefully detaching “interacting” models can cause damage to the Velcro and Styrofoam balls. However, in our implementations of the activity, only a couple of models were slightly damaged and were easily repaired through reinforcement with super glue. It is important for instructors to discuss the role and limitations of the models such as the Velcro to simulate strength of IMFs while the glue on the “atoms” relate to the difference in strength of the covalent bonds in comparison. The models do not provide distance relationships, like some general chemistry texts. Another limitation in our implementation of the activity was a time limit on each box due to the limited number of boxes (three). Because the students were limited on the time (~10 min) for which they had exposure to the boxes, they would not have been able to go back over their answers for very long if there was doubt about the comparison in strength from one model to another. It may be helpful to have a discussion or review with the instructor using the models. Some students had some prior knowledge of intermolecular attractions; how their conceptions of terminology changed was not assessed specifically.

Additionally, some students seemed to have a tendency to explain the physical properties with the simplest answers such as stating that H-bonding is stronger than dipole–dipole forces. When the explanation was not this simple, and the answers would require a more conceptual understanding or the use of the models (as in our postactivity), we believe that some students avoided answering the question in some cases. Examples of this avoidance are not answering the question, restating definitions, or simply saying the shape is different. This may suggest some cognitive dissonance for which follow up questions or interviews would be helpful to evaluate. Another change that can be made is having alcohols of various chain lengths for comparison of numerous IMFs vs strength of individual IMFs. Alternatively, an additional box may be added that contains isomers of butane, pentane, or hexane for comparison of the shapes of those isomers vs number of carbons. It should be noted that these models are not intended to eliminate all misconceptions of IMFs but rather provide a starting point for instructors to provide students with models of the microscopic level. There are data in the [Supporting Information](#); however, these data are not meant to show explicit learning gains because the purpose of the preactivity is for instructors to see what students knew prior to the modeling activity as a starting point for discussion. Examples of extensions and misconceptions not addressed by the models are discussed in the [Supporting Information](#).

CONCLUSION

Here, we present an activity for teaching or reviewing intermolecular forces in college chemistry classrooms. The activity incorporates inexpensive materials and multisensory learning. While some commercially available model kits may be very durable, such as the magnetic Water Kit water molecules from 3D Molecular Designs,²⁷ these kits can be expensive and limited in versatility. Our models can be used to demonstrate and teach all types of intermolecular forces through tactile learning; therefore, it has potential for use as a teaching tool for students who learn best through “hands-on” activities on a wide scale. Our preliminary implementation of

this activity has allowed us to assess the likeability of the activity and to observe potential student understanding regarding intermolecular forces at the microscopic level. In addition, this study may act as a stepping stone to future studies to evaluate the activity’s effectiveness in helping students form their mental models on IMFs.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available at <https://pubs.acs.org/doi/10.1021/acs.jchemed.0c00460>.

Student preactivity, in-class activity, and postactivity with instructor notes that include discussion points, modifications to the activities or model misconceptions to address, and examples of student responses ([PDF](#), [DOCX](#))

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Notes

The authors declare no competing financial interest.

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